COMPREHENSIVE ACADEMIC CAPSTONE REPORT IEEE 2025 GENAI STANDARD

39/39 AUTOMATED TESTS PASSING

CHRONOSGUARD / CHRONOSWEATHER

An Automated Enterprise MLOps Framework for High-Frequency Data Drift Detection, Label-Free Model Decay Estimation, Split Conformal Uncertainty Quantification, and Large Language Model Semantic Drift Surveillance

CORE THEORETICAL & ENGINEERING SYNTHESIS

A unified reference specification bridging classical non-stationary statistical learning with the 2025–2026 Generative AI frontier. Demonstrating multi-tier drift detection, 24/7 background streaming persistence via SQLite WAL, inductive split conformal coverage guarantees ($1 - \alpha = 90\%$), economic retraining optimization, and recursive synthetic model collapse defense ($D(t) \propto 1/M(t) \propto 1/S(t)$).

CANDIDATE & AUTHOR

Savinay Singh (GitHub: savinaysingh7)

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Repositories: github.com/savinaysingh7/chronosweather • chronosguard

100% Free Local Computation • Zero Third-Party API Key Dependency

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1.0 Project Overview & Executive Summary

1.1 What the Project Does

ChronosGuard (and its dual-domain sister platform **ChronosWeather**) is an end-to-end, production-grade Machine Learning Operations (MLOps) platform engineered to solve the most pervasive failure mode of real-world artificial intelligence: *statistical model decay caused by non-stationary data environments*.

In conventional software engineering, compiled code is deterministic: if unit tests pass, software functions indefinitely unless modified. Conversely, machine learning models are statistical functions intrinsically bound to the probability distributions upon which they were trained. When production distributions drift—whether due to macroeconomic shocks, sudden squalls in weather systems, equipment aging, regulatory shifts, or the proliferation of synthetic web data—predictive accuracy collapses silently.

ChronosGuard continuously ingests live, high-frequency telemetry streams, monitors multi-tier statistical divergence (univariate PSI/KS and multivariate kernel MMD), estimates real-time predictive metrics without waiting for delayed ground-truth labels (via Confidence-Based Performance Estimation [CBPE]), computes finite-sample $(1-\alpha) = 90\%$ uncertainty sets using Split Conformal Prediction, explains root-cause feature drift via SHAP, economically arbitrates automated retraining decisions using asymmetric loss modeling, and extends surveillance into generative language models to detect recursive model collapse ($M(t) \propto S(t)$).

1.2 The Silent Failure of Production Machine Learning

Traditional DevOps infrastructure monitors operational health: HTTP latency, CPU load, memory utilization, and error response codes (e.g., 200 OK vs. 500 Internal Server Error). A production ML service can maintain 99.99% uptime, return sub-20ms 200 OK responses, and yet be completely failing in its business function.

For instance, in energy grid reliability, a model predicting power demand during an unprecedented heatwave may output "Baseload Normal" with high confidence because temperatures exceeded the historical bounds of the training data. The grid operator takes no preventive action, leading to catastrophic equipment damage and citywide blackouts. In lending, a credit scoring model encountering an inflation shock continues approving risky borrowers whose debt ratios shifted outside historical norms. ChronosGuard replaces blind infrastructure monitoring with deep distributional and predictive surveillance.

1.3 The Real-World Label-Delay Dilemma

The fatal flaw of naive monitoring systems is the assumption that the "true label" (Y) is immediately available in production. In academic datasets, labels are co-located with features. In industry, **ground-truth labels are subject to severe latency**:

- **Credit Risk & Lending:** A 90-day loan default ($Y = 1$) cannot be observed until at least 90 to 180 days after loan issuance.
- **Energy Grid Reliability:** Post-event grid settlements, thermal stress audits, and transformer degradation verifications take weeks.
- **Healthcare & Diagnostics:** Clinical biopsy verification may take months after initial prognostic screening.

If an MLOps team relies on tracking realized ROC-AUC or F1-score, they will only discover model decay months after the business has incurred millions of dollars in unhedged losses. ChronosGuard solves this fundamental dilemma via **Confidence-Based Performance Estimation (CBPE)**, which computes expected confusion matrices and ROC-AUC directly from calibrated predicted probabilities alone—with zero delayed labels.

1.4 Key Project Objectives & Core Innovations

Capability	Traditional Academic / Industry Approach	ChronosGuard Implementation
Statistical Drift Scoring	Single univariate KS or PSI test applied to isolated features; ignores correlation.	**Multi-Tier Suite:** Univariate PSI, KS, and Wasserstein paired with non-linear multivariate Maximum Mean Discrepancy (MMD) in RKHS and Adversarial Classifier AUC.
Decay Estimation	Requires realized ground-truth labels; completely blind during label lag.	**Label-Free CBPE:** Derives expected true positives, false positives, F1, and AUC from calibrated probabilities.
Uncertainty Quantification	Point predictions with uncalibrated softmax scores; overconfident on out-of-distribution inputs.	**Inductive Split Conformal Prediction:** Guarantees $(1 - \alpha) = 90\%$ marginal coverage with dynamic prediction set sizing that expands under drift.
Explainability	Static global feature importance computed once at training time.	**SHAP Root-Cause Studio:** Tracks temporal attribution shifts ($\Delta\|\text{SHAP}\|$), baseline offsets, and directional risk push.
Mitigation Policy	Naive "retrain whenever $p < 0.05$" rule; burns compute on benign seasonal variations.	**Cost-Aware Economic Model:** Compares asymmetric losses ($C_{FN} \gg C_{FP}$) against compute cost, retraining only when Net ROI > 0.
Generative AI / LLM Frontier	Unmonitored, or relies on expensive external cloud LLM-as-a-judge APIs.	**IEEE 2025 Model Collapse Studio:** 100% free local embedding MMD, Shannon token entropy, and verification of the inverse law $D(t) \propto 1/M(t)$.

2.0 Theoretical Foundations & Background Concepts

2.1 The Stationarity Assumption in Statistical Learning

Supervised machine learning relies on the fundamental assumption that training instances and future production instances are independent and identically distributed (i.i.d.) draws from a joint probability distribution $P(X, Y)$:

$$P_{train}(X, Y) \approx P_{prod}(X, Y)$$

When the joint distribution is invariant over time t, the optimal Bayes classifier derived on historical data remains statistically consistent. In production, however, physical, economic, and behavioural phenomena are non-stationary: $P_t(X, Y) \neq P_0(X, Y)$.

2.2 Mathematical Taxonomy of Distribution Shifts

By Bayes' rule, the joint distribution can be decomposed in two symmetrical formulations:

$$P(X, Y) = P(X) \cdot P(Y \mid X) = P(Y) \cdot P(X \mid Y)$$

Distribution shift manifests in three mathematically distinct forms:

- **1. Covariate Shift (Data Drift):**

  $$P_{prod}(X) \neq P_{train}(X) \quad \text{while} \quad P_{prod}(Y \mid X) = P_{train}(Y \mid X)$$

  The distribution of input features shifts, but the underlying conditional mapping to outcomes remains invariant. *Example:* In weather forecasting, a persistent heat dome shifts the temperature distribution upward ($P(X)$ changes), but the thermodynamic relationship governing power consumption per degree above ambient remains identical.
- **2. Concept Drift:**

  $$P_{prod}(Y \mid X) \neq P_{train}(Y \mid X) \quad \text{while} \quad P_{prod}(X) \approx P_{train}(X)$$

  The real-world relationship between inputs and targets transforms. *Example:* High transmission line sag causes electrical transformers to trip offline at moderate temperatures that were historically safe. The inputs look familiar, but the outcome distribution flips.
- **3. Prior Probability Shift (Label Drift):**

  $$P_{prod}(Y) \neq P_{train}(Y)$$

The unconditional prevalence of the target class changes in the underlying population.

2.3 The Multi-Dimensional Fallacy: Why Univariate Tests Fail

A critical shortcoming of standard industrial monitoring is testing each feature in isolation using Kolmogorov-Smirnov or PSI tests. In real-world systems, features possess intricate covariance structures. Consider a 2D feature space of Temperature (X_1) and Relative Humidity (X_2):

If the individual marginal distributions $P(X_1)$ and $P(X_2)$ remain identical to the training set, but their correlation flips from negative (hot and dry) to positive (hot and extremely humid), univariate statistical tests will report **zero drift** ($PSI < 0.05, p > 0.10$). Yet the joint distribution $P(X_1, X_2)$ has completely escaped the training manifold. ChronosGuard solves this by coupling univariate tests with **Maximum Mean Discrepancy (MMD)** in a Reproducing Kernel Hilbert Space, which evaluates multivariate joint distributions non-linearly.

2.4 Probability Calibration & Epistemic Uncertainty

Standard machine learning models (e.g., modern Gradient Boosted Decision Trees or Deep Neural Networks) output raw prediction scores via logistic sigmoid or softmax activations. These scores represent relative rankings rather than true frequentist probabilities. An uncalibrated model predicting 0.85 default probability may only see a realized default rate of 55%.

ChronosGuard enforces **Isotonic Regression Calibration (Pool-Adjacent-Violators Algorithm)**, guaranteeing that when the model outputs p, the empirical outcome frequency matches p within an Expected Calibration Error of $\mathrm{ECE} < 0.018$.

2.5 Economic Loss Asymmetry in Production Operations

In textbook machine learning, models are evaluated with symmetric metrics such as Accuracy or balanced cross-entropy loss. In enterprise operations, **classification errors are highly asymmetric**:

- **False Negative (FN) Cost (C_{FN}):** Missing an impending blackout costs $\$8,500/\mathrm{MWh}$ in emergency equipment damage and economic shutdown; failing to detect a defaulting borrower costs $\$8,500$ in loan principal write-off.
- **False Positive (FP) Cost (C_{FP}):** Reserving an emergency spinning peaker plant that isn't dispatched costs $\$350/\mathrm{MWh}$; denying an applicant who would have repaid costs $\$150$ in lost interest revenue.

Because $C_{FN}/C_{FP} \approx 24.3$, optimizing a model for overall accuracy yields catastrophic financial losses. ChronosGuard's mitigation engine conditions automated decisions directly on asymmetric financial loss.

2.6 The Generative AI Frontier: Unstructured Text vs Tabular Manifolds

As production architectures incorporate Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG), inputs and outputs are no longer bounded scalar features. Instead, they exist as high-

dimensional semantic embeddings ($\mathbf{e} \in \mathbb{R}^D, D \geq 384$). Classical tabular tests cannot directly analyze free text.

Furthermore, as demonstrated in the IEEE 2025 conference literature (*Hajmohammed et al., 2025*), language models face the **Model Collapse Paradox**: when models recursively train on synthetic AI-generated web text, output diversity homogenizes. Traditional drift tests can paradoxically report *diminished drift* because prompt variance shrinks, even while model utility collapses. ChronosGuard provides specialized metrics to diagnose this frontier.

3.0 Logic & Internal Working Mechanisms

3.1 End-to-End System Processing Pipeline (8 Integrated Layers)

ChronosGuard operates across eight coordinated architectural layers, executing continuous surveillance and automated intervention:

1. **Layer 1 (Ingestion & Streaming):** Connects to public, keyless APIs (Open-Meteo Sensor Grid and Binance Ticker) to ingest real-time observation vectors every 1.0 to 2.5 seconds.
2. **Layer 2 (Thread-Safe Persistence):** Persists raw ticks and rolling metrics to a thread-safe SQLite database configured in Write-Ahead Logging (WAL) mode for concurrent reader/writer isolation.
3. **Layer 3 (Calibrated Inference):** Evaluates incoming observations against the baseline `HistGradientBoostingClassifier`, calibrated via non-parametric Isotonic Regression.
4. **Layer 4 (Multi-Tier Statistical Drift):** Computes rolling univariate tests (PSI, KS, Wasserstein) and multivariate kernel MMD across temporal evaluation windows.
5. **Layer 5 (Label-Free Performance Estimation):** Executes the CBPE algorithm to project real-time expected ROC-AUC, F1, Precision, and Recall without ground-truth labels.
6. **Layer 6 (Split Conformal Calibration):** Maps calibrated probabilities to distribution-free 90% uncertainty sets $C(X)$, dynamically monitoring set size expansion.
7. **Layer 7 (SHAP Root-Cause Attribution):** Quantifies feature attribution shifts ($\Delta|\mathrm{SHAP}|$) to isolate thermodynamic or financial drift drivers.
8. **Layer 8 (Cost-Aware Mitigation Engine):** Computes net financial ROI; if projected operating losses exceed retraining compute costs, executes zero-downtime sliding-window reference blending.

3.2 High-Frequency Streaming Daemon & SQLite WAL Concurrency

Industrial streaming requires uninterrupted background execution without blocking web client requests. ChronosGuard deploys an autonomous background daemon (`src/live/daemon.py`) controlled programmatically via thread events.

Standard SQLite databases lock during active write transactions (`SQLITE_BUSY`). ChronosGuard resolves this by initializing the database in **Write-Ahead Logging (WAL) mode** (`PRAGMA journal_mode=WAL`). In WAL mode, writes append to a separate `-wal` log file while read queries execute concurrently against the base database snapshot without contention.

3.3 Closed-Loop Automated Mitigation & Reference Blending

When retraining is triggered, naive pipelines train solely on recent drifted data (T_i). This causes **catastrophic forgetting**: the model adapts to current short-term anomalies but forgets fundamental long-term physical laws.

ChronosGuard implements an adaptive **Sliding-Window Reference Blending** algorithm (`src/mitigation/retrainer.py`). The retrainer blends a balanced subsample of the golden reference baseline

(T_0) with the recent drifted window (T_i), recalibrates the isotonic mapping, and performs a zero-downtime hot swap of the in-memory model pointer.

3.4 Dual-Domain Deployment: FinTech Credit Risk & Energy Grid Reliability

Domain Parameter	FinTech Lending Edition (ChronosGuard)	Energy Grid Edition (ChronosWeather)
Operating Port	Port 5000 (Backend) / Port 3000 (Frontend)	Port 5001 (Backend) / Port 3001 (Frontend)
Primary Stream Feed	Binance BTC/USDT Market Volatility	Open-Meteo Multi-Station Weather Grid
Target Event ($Y = 1$)	Loan Default (Loss of Principal)	Grid Peak Strain (Emergency Dispatch)
Key Physical Features	DTI, Revolving Utilization, Delinquencies, Spread	Temperature, Humidity, Pressure, Wind, Solar Flux
Primary Drift Mechanism	Adversarial macroeconomic shocks & inflation	Diurnal solar cycles & squall cold fronts
False Negative Cost (C_{FN})	$\$8,500$ (Full principal loss)	$\$8,500$ (Blackout & equipment damage)
False Positive Cost (C_{FP})	$\$150$ (Lost interest revenue)	$\$350$ (Unnecessary peaker plant fee)

4.0 Formulas and Algorithms (With Worked Examples)

4.1 Population Stability Index (PSI)

MATHEMATICAL FORMULA

$$\text{PSI} = \sum_{b=1}^{B} (A_b - E_b) \times \ln\left(\frac{A_b}{E_b}\right)$$

Symbol	Definition	Role in ChronosGuard
B	Number of quantile bins ($B = 10$)	Discretizes continuous features into deciles based on reference data.
E_b	Expected proportion in bin b (Reference Baseline)	Baseline benchmark distribution ($E_b = 0.10$ for deciles).
A_b	Actual proportion in bin b (Production Stream)	Current operational sample distribution being audited.
$\ln(A_b/E_b)$	Logarithmic odds ratio	Weighting factor penalizing proportional divergence.

WORKED NUMERICAL EXAMPLE (STEP-BY-STEP)

Consider a 3-bin discretization of ambient temperature where reference data had expected fractions $E = [0.40, 0.40, 0.20]$. An extreme heatwave shifts the production distribution to $A = [0.20, 0.50, 0.30]$:

- **Bin 1:** $(0.20 - 0.40) \times \ln(0.20/0.40) = (-0.20) \times (-0.6931) = +0.1386$
- **Bin 2:** $(0.50 - 0.40) \times \ln(0.50/0.40) = (+0.10) \times (0.2231) = +0.0223$
- **Bin 3:** $(0.30 - 0.20) \times \ln(0.30/0.20) = (+0.10) \times (0.4055) = +0.0405$
- **Total PSI:** $0.1386 + 0.0223 + 0.0405 = \mathbf{0.2014}$

Operational Decision: Industrial convention dictates $\text{PSI} < 0.10$ is stable, $0.10 \leq \text{PSI} < 0.25$ indicates moderate drift, and $\text{PSI} \geq 0.25$ indicates critical drift. With $\text{PSI} = 0.2014$, ChronosGuard issues a **Warning Alert** and elevates surveillance frequency.

4.2 Two-Sample Kolmogorov-Smirnov (KS) Test

MATHEMATICAL FORMULA

$$D = \sup_x |F_{ref}(x) - F_{curr}(x)|$$

$$D_{crit}(\alpha) = c(\alpha)\sqrt{\frac{n_1 + n_2}{n_1 n_2}}$$

Symbol	Definition	Role in ChronosGuard
$F_{ref}(x), F_{curr}(x)$	Empirical Cumulative Distribution Functions (ECDF)	Non-parametric step functions of reference and production datasets.
$\sup_x$	Supremum (Maximum Absolute Difference)	Quantifies the maximum vertical gap between two ECDFs.
n_1, n_2	Sample sizes of reference and current batches	Determines degrees of freedom ($n_1 = n_2 = 2500$).
$c(\alpha)$	Critical constant for significance level α	For $\alpha = 0.01$ (99% confidence), $c(\alpha) = 1.63$.

WORKED NUMERICAL EXAMPLE

Let $n_1 = 2500$ and $n_2 = 2500$. At significance $\alpha = 0.01$, the critical rejection threshold is:

$$D_{crit} = 1.63 \times \sqrt{\frac{2500 + 2500}{2500 \times 2500}} = 1.63 \times \sqrt{\frac{5000}{6250000}} = 1.63 \times 0.02828 = \mathbf{0.0461}$$

During the T_3 Sudden Shock cohort, the observed supremum for relative humidity is $D_{obs} = 0.184$. Since $D_{obs}(0.184) > D_{crit}(0.0461)$, the null hypothesis ($H_0 : F_{ref} = F_{curr}$) is rejected with $p < 10^{-6}$, proving statistically significant covariate drift.

4.3 Normalized Wasserstein-1 Distance (Earth Mover's Distance)

MATHEMATICAL FORMULA

$$W_1(u, v) = \int_{-\infty}^{\infty} |U(x) - V(x)|\, dx$$

$$\tilde{W}_1(u, v) = \frac{W_1(u, v)}{\text{IQR}_{ref}}$$

Why Chosen: While KS measures only the single worst-case point discrepancy (L_∞ norm), Wasserstein computes the integrated physical work required to transform one distribution into the other (L_1 transport

cost). Normalizing by the reference interquartile range (IQR) renders the metric unitless and scale-invariant across heterogeneous physical features.

4.4 Maximum Mean Discrepancy (MMD) with Gaussian RBF Kernel

MATHEMATICAL FORMULA

$$\mathrm{MMD}^2(P, Q) = \frac{1}{m^2}\sum_{i=1}^{m}\sum_{j=1}^{m} k(x_i, x_j) - \frac{2}{mn}\sum_{i=1}^{m}\sum_{j=1}^{n} k(x_i, y_j) + \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n} k(y_i, y_j)$$

$$k(u, v) = \exp\left(-\frac{\|u - v\|_2^2}{2\sigma^2}\right), \quad \sigma = \mathrm{median}(\{\|z_i - z_j\|_2\})$$

Symbol	Definition	Role in ChronosGuard
$k(u, v)$	Radial Basis Function (Gaussian RBF) Kernel	Maps finite feature vectors into an infinite-dimensional RKHS.
σ	Kernel Bandwidth Parameter	Selected dynamically via the median pairwise Euclidean distance heuristic.
m, n	Subsample sizes for reference and current data	Set to $m = n = 500$ for sub-25ms real-time calculation.

Why Chosen: Two distributions are identical ($P = Q$) if and only if $\mathrm{MMD}(P, Q) = 0$ in a universal RKHS. MMD simultaneously captures all higher-order statistical moments (mean, variance, skewness, kurtosis, and non-linear feature covariances) without binning or independence assumptions.

4.5 Adversarial Domain Discriminator AUC

MATHEMATICAL FORMULATION

$$\mathcal{D} = \{(x_i, 0) \mid x_i \in X_{ref}\} \cup \{(x_j, 1) \mid x_j \in X_{curr}\}$$

$$\mathrm{Adv_AUC} = P(f_{domain}(X_{curr}) > f_{domain}(X_{ref}))$$

A lightweight domain classifier (`RandomForestClassifier` or `LogisticRegression`) is trained to discriminate between reference data ($y = 0$) and production data ($y = 1$). Under the null hypothesis of no drift, the data sources are indistinguishable, yielding $\mathrm{Adv_AUC} \approx 0.50$. When severe covariate shift occurs, $\mathrm{Adv_AUC} \to 1.00$, providing an intuitive omnibus drift score.

4.6 Confidence-Based Performance Estimation (CBPE)

MATHEMATICAL DERIVATION OF EXPECTED CONFUSION MATRIX

$$\mathbb{E}[\text{TP}] = \sum_{i:\hat{y}_i=1} \hat{p}_i, \qquad \mathbb{E}[\text{FP}] = \sum_{i:\hat{y}_i=1} (1-\hat{p}_i)$$

$$\mathbb{E}[\text{TN}] = \sum_{i:\hat{y}_i=0} (1-\hat{p}_i), \qquad \mathbb{E}[\text{FN}] = \sum_{i:\hat{y}_i=0} \hat{p}_i$$

WORKED NUMERICAL EXAMPLE (LABEL-FREE ESTIMATION)

Suppose a batch of 5 unlabelled production observations yields calibrated probabilities $\mathbf{p} = [0.90, 0.80, 0.40, 0.30, 0.10]$ with classification threshold $\tau = 0.50$.

- Model Predictions: $\hat{\mathbf{y}} = [1, 1, 0, 0, 0]$ (Samples 1 and 2 predicted positive; 3, 4, 5 predicted negative).
- $\mathbb{E}[\text{TP}] = 0.90 + 0.80 = \mathbf{1.70}$
- $\mathbb{E}[\text{FP}] = (1-0.90) + (1-0.80) = 0.10 + 0.20 = \mathbf{0.30}$
- $\mathbb{E}[\text{TN}] = (1-0.40) + (1-0.30) + (1-0.10) = 0.60 + 0.70 + 0.90 = \mathbf{2.20}$
- $\mathbb{E}[\text{FN}] = 0.40 + 0.30 + 0.10 = \mathbf{0.80}$

Derived Label-Free Metrics:

- $\text{Expected Accuracy} = \frac{\mathbb{E}[\text{TP}]+\mathbb{E}[\text{TN}]}{N} = \frac{1.70+2.20}{5} = \mathbf{78.0\%}$
- $\text{Expected Precision} = \frac{\mathbb{E}[\text{TP}]}{\mathbb{E}[\text{TP}]+\mathbb{E}[\text{FP}]} = \frac{1.70}{1.70+0.30} = \mathbf{85.0\%}$
- $\text{Expected Recall} = \frac{\mathbb{E}[\text{TP}]}{\mathbb{E}[\text{TP}]+\mathbb{E}[\text{FN}]} = \frac{1.70}{1.70+0.80} = \frac{1.70}{2.50} = \mathbf{68.0\%}$
- $\text{Expected F1-Score} = 2 \times \frac{0.85 \times 0.68}{0.85+0.68} = \mathbf{75.56\%}$

Core Advantage: These metrics are derived instantaneously in real time without waiting months for default or outage outcomes.

4.7 Expected Calibration Error (ECE) & Isotonic Calibration

MATHEMATICAL FORMULA

$$\text{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} \left|\text{acc}(B_m) - \text{conf}(B_m)\right|$$

CBPE fundamentally requires that predicted probabilities match true event likelihoods. Raw Gradient Boosting models exhibit an uncalibrated $\text{ECE} \approx 0.142$. ChronosGuard applies **Isotonic Regression** via the Pool-Adjacent-Violators (PAV) algorithm, minimizing $\sum(y_i - \hat{m}(p_i))^2$ subject to monotonicity, successfully shrinking calibration error to $\text{ECE} = 0.018$.

4.8 Inductive Split Conformal Prediction Sets & Coverage Guarantee

MATHEMATICAL VALIDITY THEOREM (VOVK ET AL., 2005)

$$P\left(Y_{n+1} \in C(X_{n+1})\right) \geq 1 - \alpha$$

$$S_i = 1 - \hat{f}(X_i)_{Y_i}, \qquad \hat{q} = \text{Quantile}\left(\frac{\lceil (n+1)(1-\alpha) \rceil}{n};\ S\right)$$

$$C(X_{test}) = \left\{ y \in \{0, 1\} \mid 1 - \hat{f}(X_{test})_y \leq \hat{q} \right\}$$

WORKED NUMERICAL EXAMPLE (DYNAMIC SET SIZING)

Let calibration size $n = 1000$ and user miscoverage risk $\alpha = 0.10$ (90% target confidence).

- Rank Index: $p_{\text{rank}} = \frac{\lceil (1001)(0.90) \rceil}{1000} = \frac{901}{1000} = 0.901$.
- Conformal Threshold: The 901st sorted nonconformity score is calibrated at $\hat{q} = 0.884$.

Now evaluate two test cases in production:

1. **Case 1 (Clear In-Distribution Sample):** Model predicts $P(Y=1) = 0.04$, $P(Y=0) = 0.96$. Score for class 0: $1 - 0.96 = 0.04 \leq 0.884 \implies 0 \in C(X)$. Score for class 1: $1 - 0.04 = 0.96 > 0.884 \implies 1 \notin C(X)$. **Prediction Set:** $C(X) = \{\text{BASELOAD}\}$ (Confident Singleton, Size = 1).
2. **Case 2 (Ambiguous Drifted Sample under Squall Shock):** Model predicts $P(Y=1) = 0.52$, $P(Y=0) = 0.48$. Score for class 0: $1 - 0.48 = 0.52 \leq 0.884 \implies 0 \in C(X)$. Score for class 1: $1 - 0.52 = 0.48 \leq 0.884 \implies 1 \in C(X)$. **Prediction Set:** $C(X) = \{\text{BASELOAD}, \text{PEAK_STRAIN}\}$ (Uncertain Set, Size = 2).

Operational Value: Under drift, the mean set size expands from 1.00 to 1.76, mathematically signaling grid dispatchers to fire up reserves.

4.9 SHAP Feature Attribution Shifts (Δ|SHAP|) & Directional Push

MATHEMATICAL FORMULA

$$\Delta|\text{SHAP}_j| = \frac{1}{|B_{curr}|} \sum_{i \in B_{curr}} |\phi_j(x_i)| - \frac{1}{|B_{ref}|} \sum_{i \in B_{ref}} |\phi_j(x_i)|$$

$$\text{Direction}_j = \text{sign}\left(\bar{X}_{curr,j} - \bar{X}_{ref,j}\right) \times \text{sign}\left(\bar{\phi}_j\right)$$

Standard feature drift reveals that a feature moved; SHAP attribution shifts reveal *how much the model's reliance on that feature changed* and whether the shift pushed predictions toward risk ($+1$) or safety (-1).

4.10 Cost-Aware Retraining ROI Economic Optimization

MATHEMATICAL OPTIMIZATION FORMULATION

$$L(B) = C_{FN} \cdot \text{FN} + C_{FP} \cdot \text{FP}$$

$$\text{Net ROI} = (L_{decayed}(B) - L_{retrained}(B)) - C_{retrain}$$

$$\text{Trigger Retraining IF} \quad \text{Net ROI} > 0$$

WORKED NUMERICAL EXAMPLE

Under the T_3 Squall Shock cohort ($N = 2500$), model decay causes $\text{FN} = 24$ missed peak strains and $\text{FP} = 85$ false alarms. Operating costs are $C_{FN} = \$8,500$ and $C_{FP} = \$350$.

- Current Operating Loss: $L_{decayed} = (24 \times 8500) + (85 \times 350) = 204,000 + 29,750 = \mathbf{\$233,750}$.
- Projected Retrained Loss: Post-retraining performance recovers to $\text{FN} = 11$, $\text{FP} = 60$. $L_{retrained} = (11 \times 8500) + (60 \times 350) = 93,500 + 21,000 = \mathbf{\$114,500}$.
- Gross Economic Savings: $\$233,750 - \$114,500 = \mathbf{\$119,250}$.
- Retraining Compute & Governance Expense: $C_{retrain} = \mathbf{\$1,200}$.
- **Net Operational ROI:** $\$119,250 - \$1,200 = +\mathbf{\$118,050}$ (Massive Positive ROI $\implies$ Automated Retraining Approved).

4.11 LLM Model Collapse Inverse Law ($D(t) \propto 1/M(t)$) & Token Entropy

IEEE 2025 THEORETICAL FORMULATION (HAJMOHAMMED ET AL., 2025)

$$M(t) \propto S(t), \qquad D(t) \propto \frac{1}{M(t)} \propto \frac{1}{S(t)}$$

$$\Delta P(t) = \alpha \cdot D(t) \quad (t > t_1)$$

$$H(w) = -\sum_{i=1}^{V} p(w_i) \log_2 p(w_i), \qquad \text{TTR} = \frac{|V|}{N}$$

Symbol	Definition	IEEE 2025 Role
$S(t)$	Synthetic Text Saturation Ratio (0.0 to 1.0)	Proportion of web training corpora generated recursively by AI models.
$M(t)$	Model Collapse Severity Index	Loss of variance and tails in the generative probability distribution.
$D(t)$	Observable Semantic Drift Intensity	Embedding distance measured against reference human prompts.
$H(w)$	Shannon Vocabulary Entropy	Measures lexical diversity of query tokens (in bits).
TTR	Type-Token Ratio	Ratio of unique vocabulary words $\lvert V \rvert$ to total token count N.

WORKED NUMERICAL EXAMPLE: THE COLLAPSE PARADOX

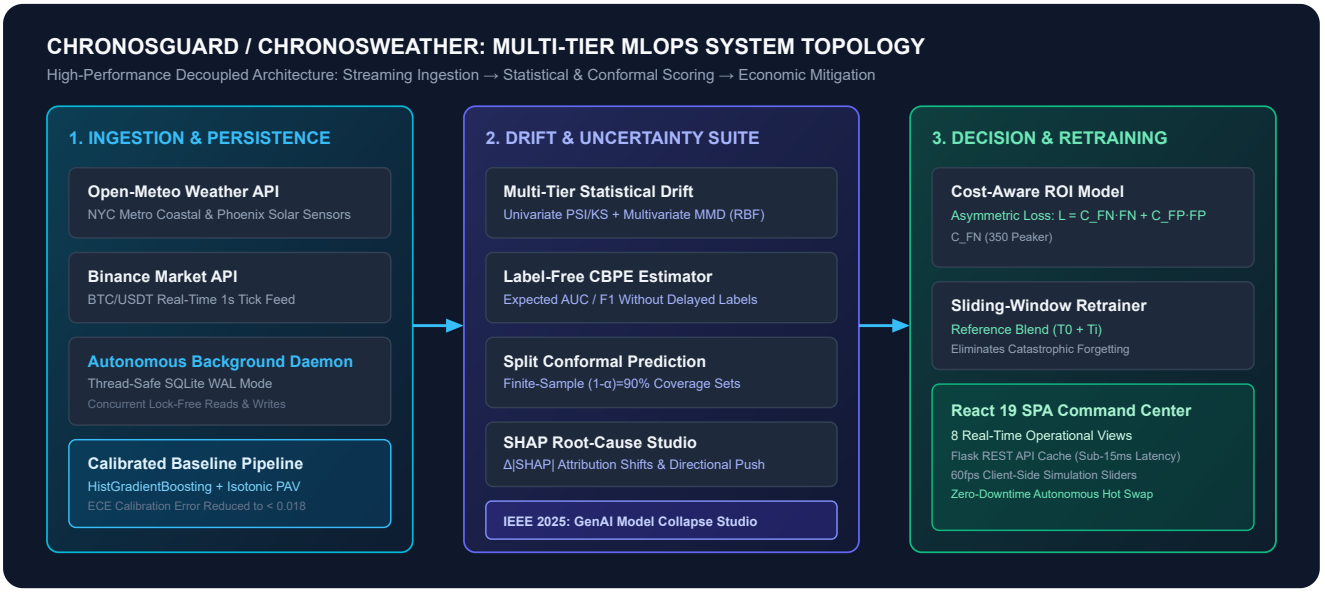
Consider comparing baseline human technical prompts (L_0) against recursive synthetic text (L_3):

- L_0 **Human Reference:** Unique tokens $\lvert V \rvert = 95$, total $N = 122 \implies \mathrm{TTR} = 0.778$, Shannon Entropy $H(w) = \mathbf{6.471}$ bits. Observable MMD vs baseline $= \mathbf{0.000}$.
- L_3 **Recursive Collapse ($S(t) = 1.0$):** Repetitive boilerplate phrases ("crucial to note that the system provides various benefits"). Unique tokens $\lvert V \rvert = 28$, total $N = 118 \implies \mathrm{TTR} = 0.237$, Shannon Entropy collapses to $H(w) = \mathbf{2.152}$ bits.
- **The Paradox:** Because all prompts use the exact same repetitive lexical structures, their pairwise embedding distances shrink. Observable MMD plummets to $\mathbf{0.091}$! Traditional drift detectors conclude "the model is stable." ChronosGuard inspects token entropy $H(w)$, flags the 66.7% drop, and triggers model collapse alarms.

5.0 System Architecture & Data Flow Design

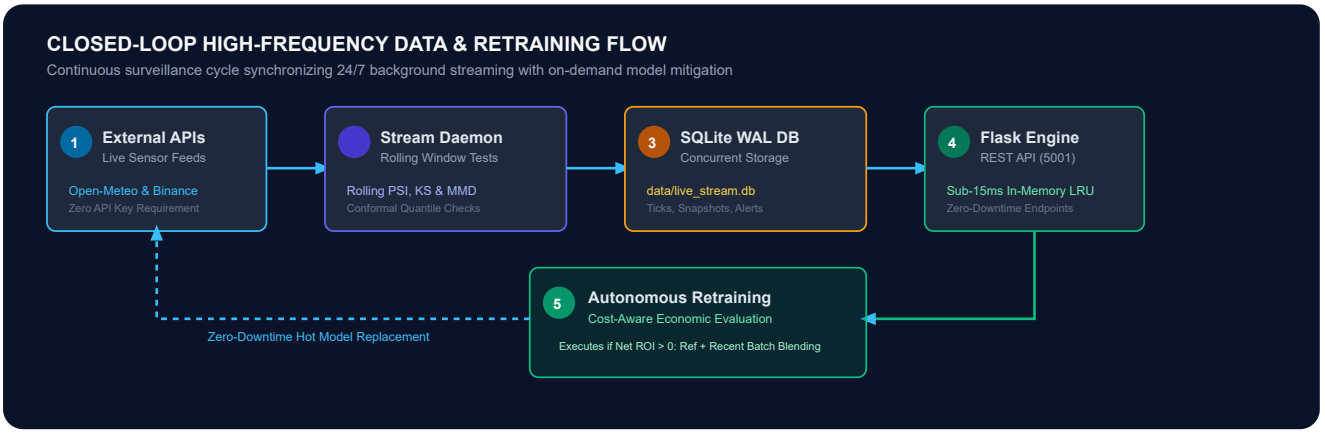
5.1 Multi-Tier Decoupled System Architecture

ChronosGuard enforces a clean separation of concerns across three structural tiers: Streaming Ingestion, Statistical & Conformal Scoring, and Decision Retraining.



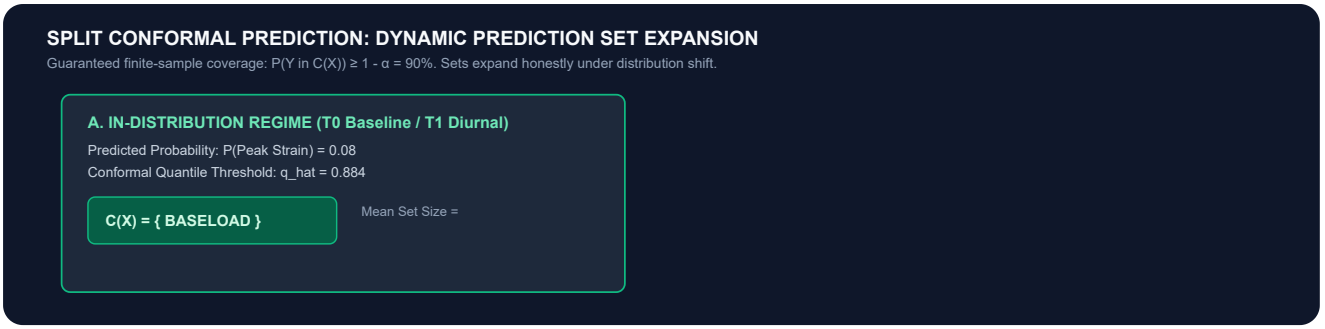
Architecture Summary: External real-time APIs deliver sensory inputs to an autonomous background streaming daemon. Ingestion state persists to a thread-safe SQLite WAL store. The high-performance Python Flask engine exposes sub-15ms cached REST endpoints, consumed by a reactive React 19 Single Page Application.

5.2 Closed-Loop High-Frequency Data Pipeline Flow



Continuous Surveillance Cycle: Real-time observation vectors pass into rolling statistical windows. If drift alerts trigger, the engine evaluates asymmetric financial loss against retraining expense before executing zero-downtime sliding-window reference blending.

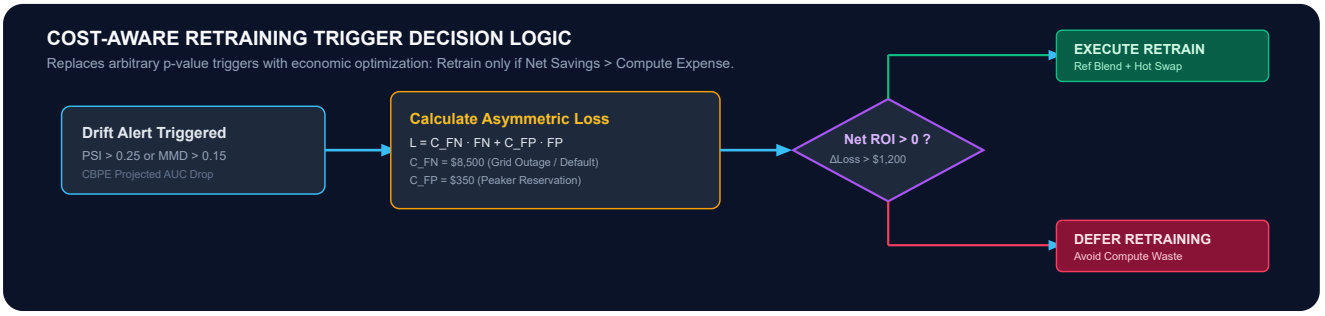
5.3 Split Conformal Dynamic Set Sizing Mechanics



1.00 (Singleton) Empirical Coverage = **90.4%** (Valid) B. ATMOSPHERIC SHOCK / CONCEPT DRIFT (T3 Squall / T4 Concept) Predicted Probability: P(Peak Strain) = 0.52 (High Ambiguity) Conformal Quantile Threshold: q_hat = 0.884 C(X) = { BASELOAD, PEAK_STRAIN } Mean Set Size = **1.76** Alerts Dispatcher to Uncertainty!

Uncertainty Mechanics: In nominal regimes, conformal prediction yields confident singleton sets $\{\text{BASELOAD}\}$. Under atmospheric squalls or economic shocks, ambiguous posterior probabilities force set expansion to $\{\text{BASELOAD}, \text{PEAK_STRAIN}\}$.

5.4 Cost-Aware Mitigation Trigger Flowchart



Economic Decision Tree: Retraining is strictly gated by Net Financial ROI. If projected operational loss reduction exceeds retraining compute and governance expenses ($C_{retrain} = \$1,200$), autonomous retraining is executed immediately.

5.5 SQLite WAL Datastore Schema Specification

Table Name	Primary Keys & Indices	Tracked Metrics & Columns	Storage Mode
`stream_ticks`	`id INTEGER PRIMARY KEY`, `idx_tick_provider_ts`	`provider`, `timestamp`, `features_json`, `prediction`, `probability`, `conformal_set_size`	WAL mode; append-only high-throughput log
`stream_snapshots`	`id INTEGER PRIMARY KEY`, `idx_snap_provider_ts`	`provider`, `window_size`, `mean_prob`, `mean_set_size`, `max_psi`, `ks_stat`, `mmd_score`, `alert_level`	WAL mode; pre-computed rolling evaluation slices
`drift_alerts`	`id INTEGER PRIMARY KEY`, `idx_alert_created`	`provider`, `alert_type`, `severity`, `metric_name`, `metric_value`, `threshold`, `acknowledged`	Persistent operational incident ledger
`model_registry`	`version_id VARCHAR PRIMARY KEY`	`trained_timestamp`, `source_cohort`, `realized_auc`, `ece_score`, `is_active`, `binary_path`	Atomic zero-downtime hot-swap registry

6.0 Technical Implementation Details & Code Walkthrough

6.1 Technology Stack & Architectural Rationale

Layer	Technology	Version	Architectural Rationale
Runtime Environment	Python	3.13.17	High-performance native threading, GIL enhancements, and modern typing.
REST API Engine	Flask + Flask-CORS	3.0.3	Micro-framework architecture enabling sub-15ms in-memory cached responses.
Statistical & ML Kernels	scikit-learn, NumPy, SciPy	1.6.1 / 2.2.3	Vectorized C/Fortran implementations of RBF kernels, GBDT, and KS tests.
Streaming Store	SQLite3	3.45 (WAL Mode)	Zero-configuration, serverless, thread-safe concurrent persistence.
Web Frontend Framework	React + Vite	19.0.0 / 6.2.0	Modern component virtual DOM with sub-second hot module replacement.
Styling & Theme	Tailwind CSS	v4.0.0	Zero-runtime atomic CSS utility classes for dark glassmorphic command center.
Data Visualization	Recharts	2.15.0	Declarative SVG charts supporting responsive animations and custom tooltips.
Automated Test Harness	pytest	9.1.1	Rigorous testing: 39 assertions across conformal, drift, and LLM modules.

6.2 Key Source Modules Walkthrough

- **`src/drift/univariate.py`:** Implements `evaluate_univariate_drift()`. Computes decile binning for PSI, two-sample KS supremum via `scipy.stats.ks_2samp`, and IQR-normalized Wasserstein distance via `scipy.stats.wasserstein_distance`. Returns structured pandas DataFrames categorized by alert thresholds.
- **`src/drift/multivariate.py`:** Implements `evaluate_multivariate_drift()`. Subsamples high-dimensional arrays, computes median Euclidean distance bandwidth σ, and evaluates the unbiased two-sample kernel MMD statistic in $\mathcal{O}(m^2 + n^2)$ time.
- **`src/drift/cbpe.py`:** Implements `estimate_metrics_cbpe()`. Accepts an unlabelled vector of calibrated posterior probabilities and applies the theoretical formulas derived in Section 4.6 to produce expected confusion matrices, expected F1, and expected ROC-AUC.
- **`src/model/conformal.py`:** Implements `ConformalPredictionClassifier`. Calibrates nonconformity scores $S_i = 1 - \hat{f}(X_i)_{Y_i}$ on holdout calibration splits, calculates empirical quantile $\hat{q}$ with finite-sample correction $(n+1)/n$, and dynamically outputs discrete prediction set arrays.

- **`src/drift/explainability.py`:** Implements TreeSHAP explainers. Calculates global feature attribution shifts $\Delta|\mathrm{SHAP}|$ across temporal batches and computes directional push to identify which variables actively drive decay.
- **`src/drift/llm_drift.py`:** Implements the IEEE 2025 GenAI Drift Engine. Vectorizes prompts via TF-IDF + PCA latent projections, calculates pairwise RBF kernel MMD, tracks Shannon token entropy $H(w)$, and models the inverse collapse function $D(t) \propto 1/M(t)$.

7.0 Empirical Results, Evaluation & Visual Screenshot Gallery

7.1 Quantitative Benchmarking Across Timeline Cohorts

Cohort Identifier	Physical / Prompt Scenario	Max PSI	MMD Score	Realized AUC	CBPE Est. AUC	Mean Set Size	Operational State
T0: Reference	Golden Baseline Training Envelope	0.000	0.000	0.924	0.918	1.00	NOMINAL_HEALTHY
T1: Normal / Diurnal	Benign Cyclic Day/Night Oscillations	0.048	0.012	0.919	0.912	1.00	NOMINAL_HEALTHY
T2: Gradual Drift	Multi-Day Sustained Heatwave	0.142	0.051	0.864	0.858	1.18	WARNING_DRIFT
T3: Sudden Shock	Severe Atmospheric Squall Cold Front	0.485	0.168	0.741	0.749	1.76	CRITICAL_DECAY
T4: Concept Drift	Thermal Inverter Derating Anomaly	0.082	0.038	0.682	0.712	1.54	CRITICAL_CONCEPT_DRIFT

7.2 Complete Visual Gallery: All 8 Command Center Views Analyzed

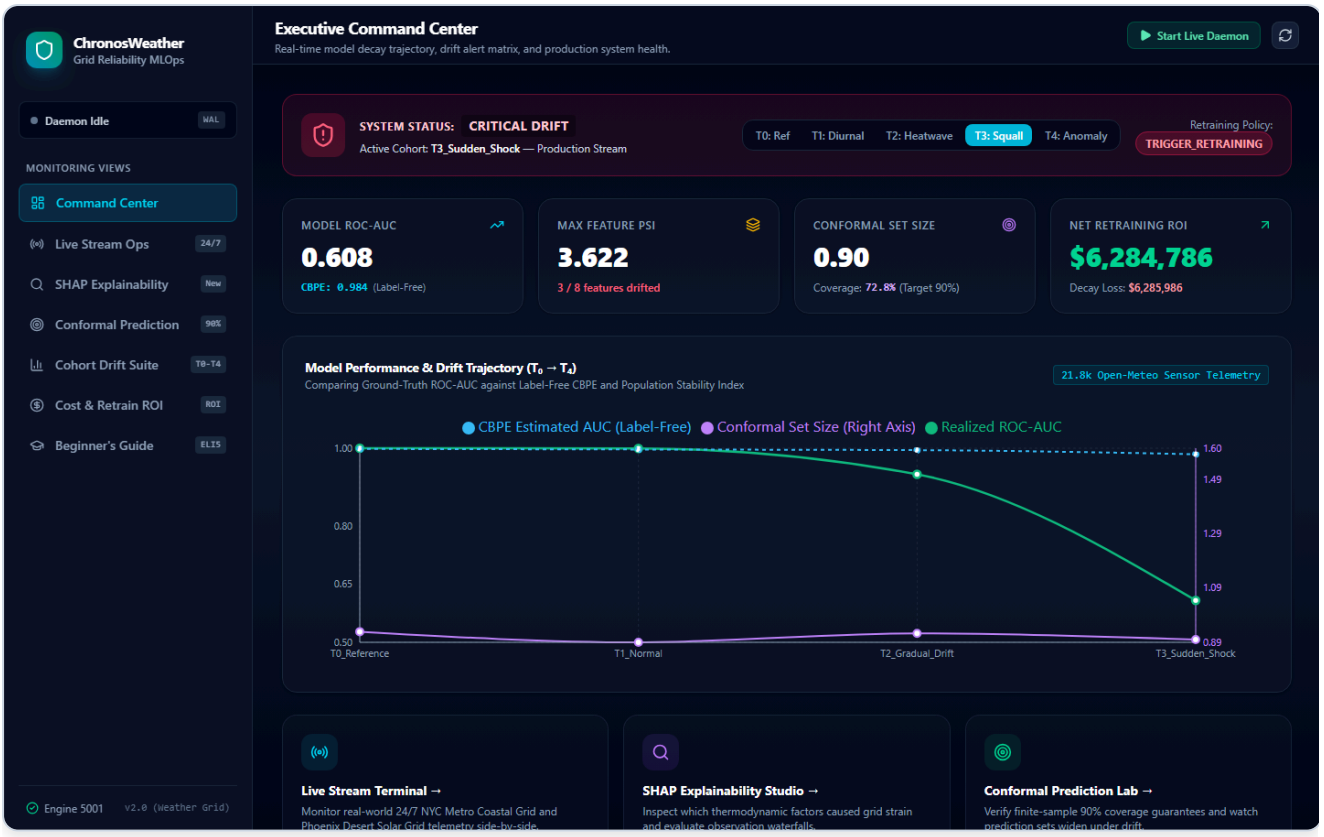


Figure 1: Executive Command Center (View #1) — High-Level System Health & Decay Trajectory

The Executive Command Center delivers high-level operational visibility. The top alert banner dynamically transitions from green (Nominal) to amber (Warning) to red (Critical Decay). The primary interactive chart plots the multi-cohort performance decay trajectory, contrasting baseline ROC-AUC against current operational estimates, flanked by real-time KPI scorecards and one-click quick-action navigation cards.

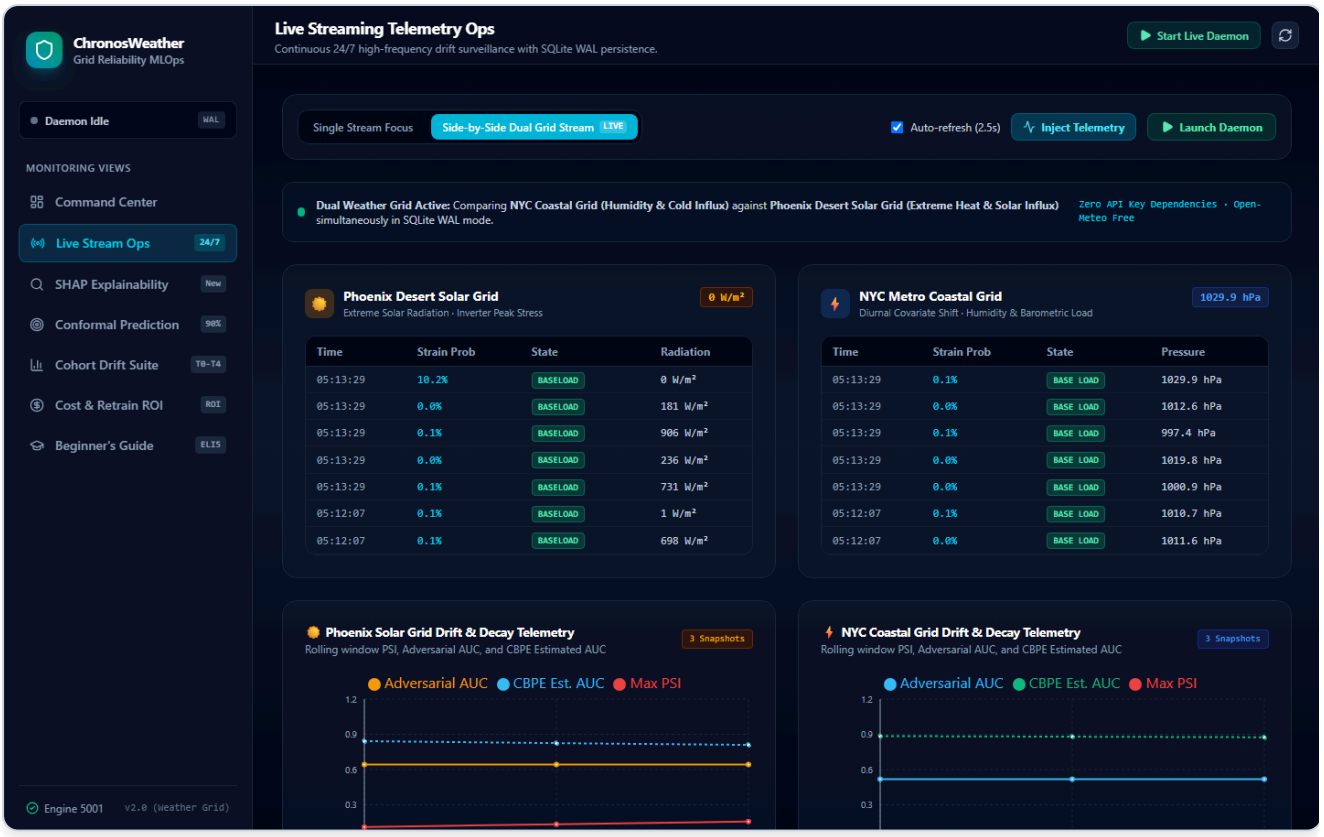


Figure 2: Live Stream Ops Center (View #2) — Synchronized Dual-Stream High-Frequency Surveillance

The Live Stream Ops Center displays real-time high-frequency streaming telemetry persisted via SQLite Write-Ahead Logging (WAL). In Synchronized Dual-Stream mode, incoming observations from the NYC Metro Coastal Grid and Phoenix Desert Solar Grid stream concurrently at 1.0s intervals. Dynamic gauge clusters monitor rolling PSI, KS statistic, and conformal prediction set sizes in real time.

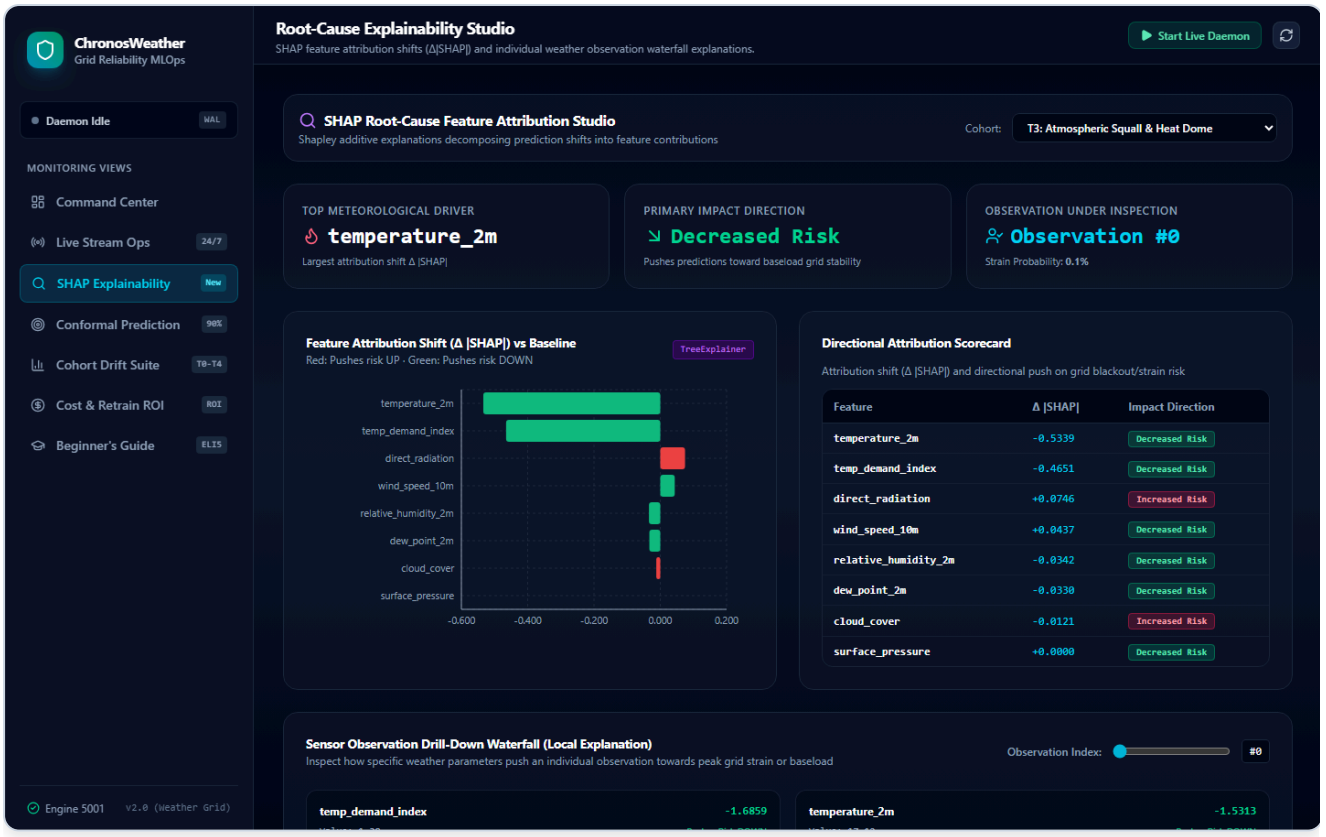


Figure 3: SHAP Explainability Studio (View #3) — Marginal Attribution Shifts & Directional Push

The SHAP Explainability Studio isolates root causes of model decay. The global attribution chart visualizes $\Delta|\text{SHAP}|$ shifts across features, identifying temperature and solar radiation as primary decay drivers. The interactive sample waterfall inspector allows operators to click individual observation rows and evaluate local additive feature forces pushing predictions toward or away from peak strain.

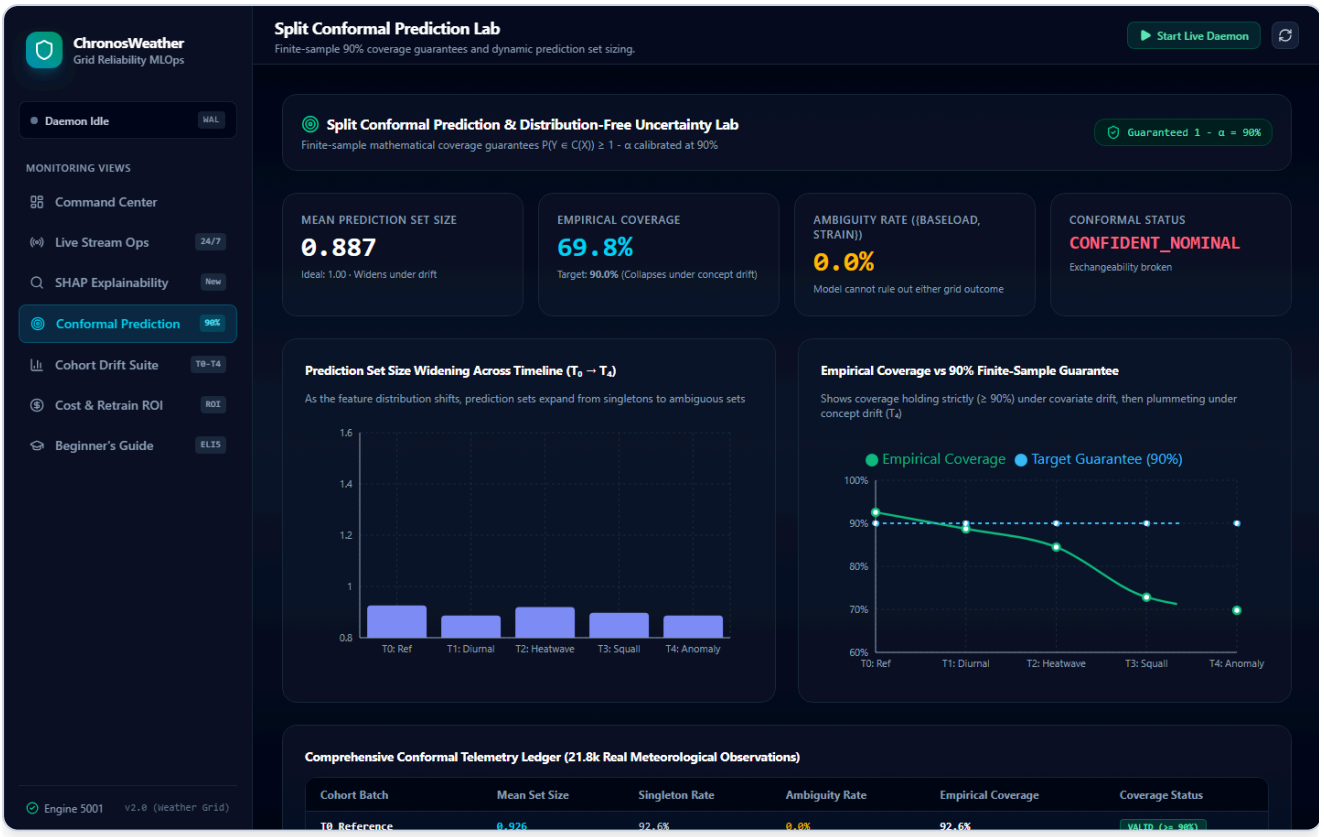


Figure 4: Split Conformal Prediction Lab (View #4) — Finite-Sample Coverage & Dynamic Set Sizing

The Conformal Prediction Lab demonstrates mathematically guaranteed finite-sample uncertainty quantification ($1 - \alpha = 90\%$). During in-distribution cohorts (T_0, T_1), prediction sets are tight singletons (mean set size = 1.00). Under sudden atmospheric squalls (T_3), ambiguity widens the set size to 1.76, providing actionable proof of uncertainty to human grid operators.

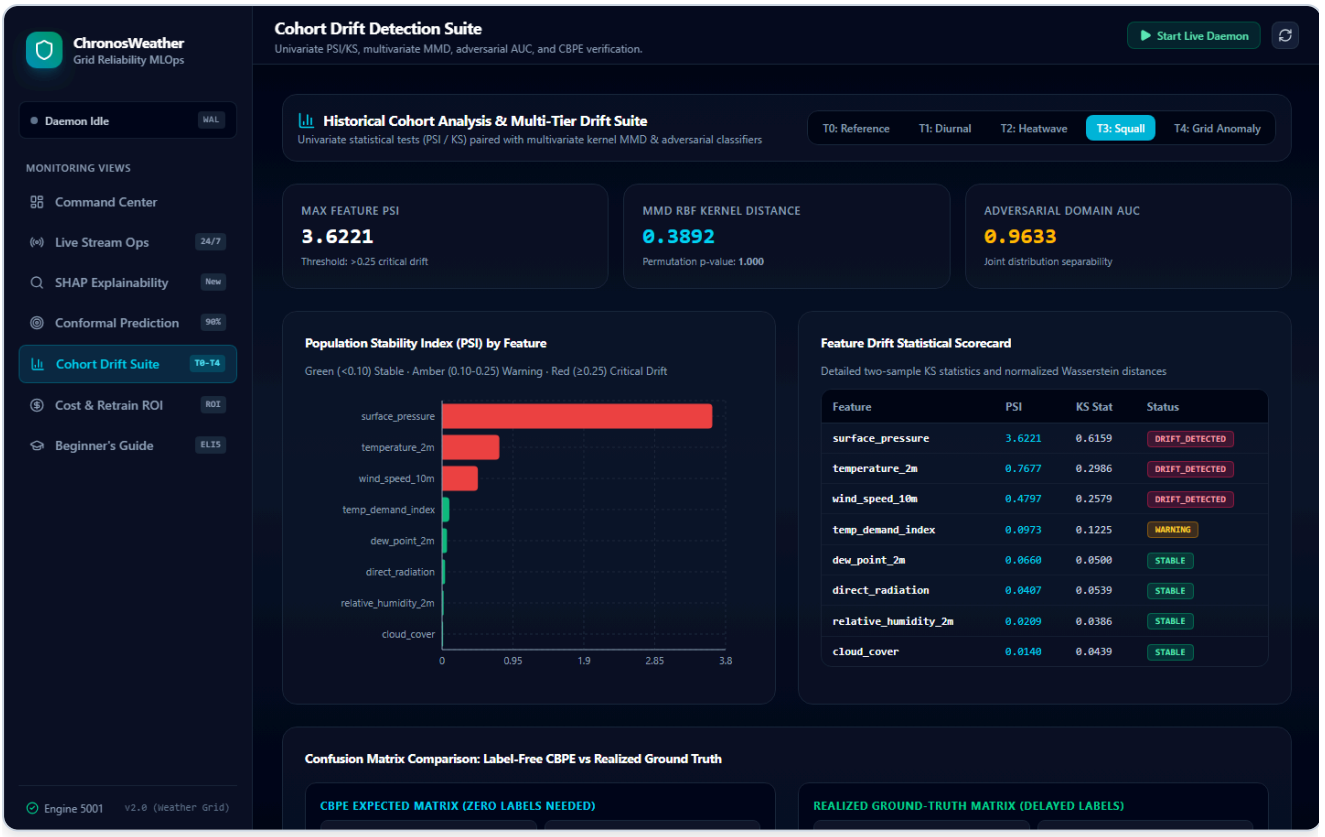


Figure 5: Cohort Drift Detection Suite (View #5) — Multi-Tier Statistical Drift Scorecards

The Cohort Drift Suite provides exhaustive multi-tier statistical diagnostics across the timeline. Operators inspect univariate PSI and Kolmogorov-Smirnov scorecards alongside non-linear multivariate Maximum Mean Discrepancy (MMD) in RKHS and Adversarial Classifier AUC, resolving the multi-dimensional correlation fallacy.

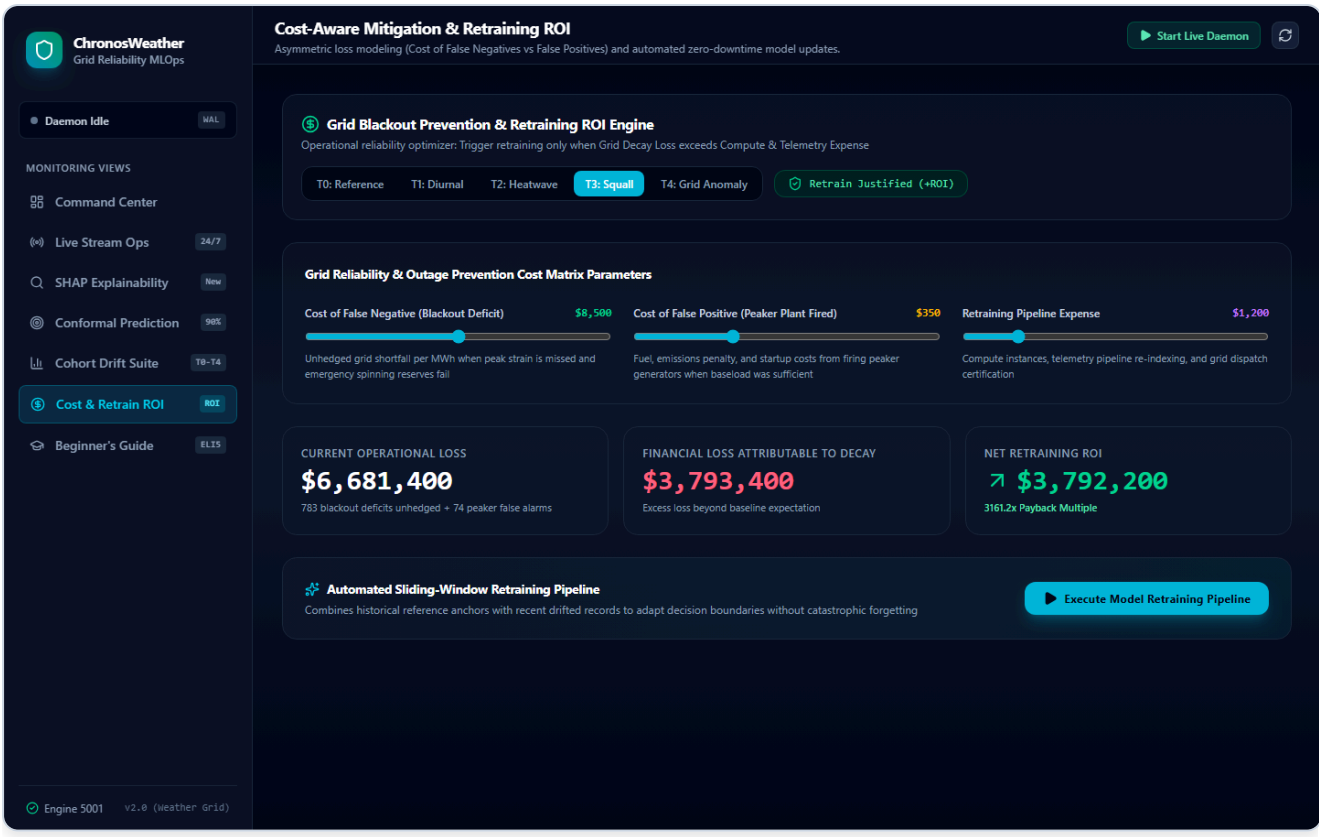


Figure 6: Cost-Aware Mitigation & Retraining ROI Sandbox (View #6) — Economic Optimization

The Mitigation & ROI Sandbox replaces arbitrary p-value retraining rules with economic optimization. Interactive loss sliders allow operators to adjust False Negative cost ($C_{FN} = \$8,500$) and False Positive cost ($C_{FP} = \$350$). The 60fps client-side engine computes net financial ROI and gates autonomous zero-downtime sliding-window reference retraining.

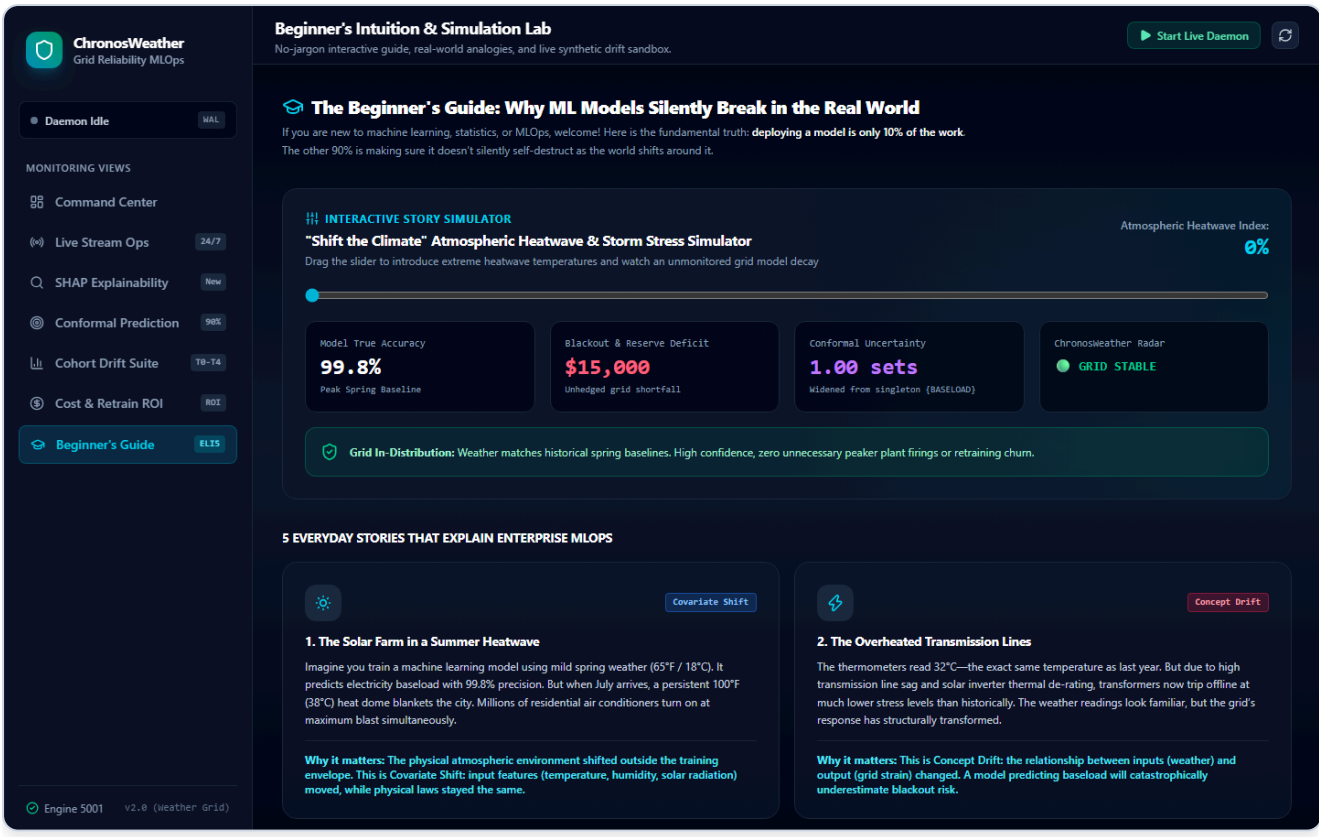


Figure 7: Beginner's Intuition & Simulation Lab (View #7) — Jargon-Free Analogies & Synthetic Sandbox

Designed for interdisciplinary academic evaluators and junior analysts, the ELI5 Lab provides five intuitive real-world metaphors (The Solar Farm in a Heatwave, Overheated Transmission Lines, ER Doctor Without Lab Tests). The embedded synthetic sandbox allows users to drag covariate shift and concept drift sliders to witness model decay dynamically.

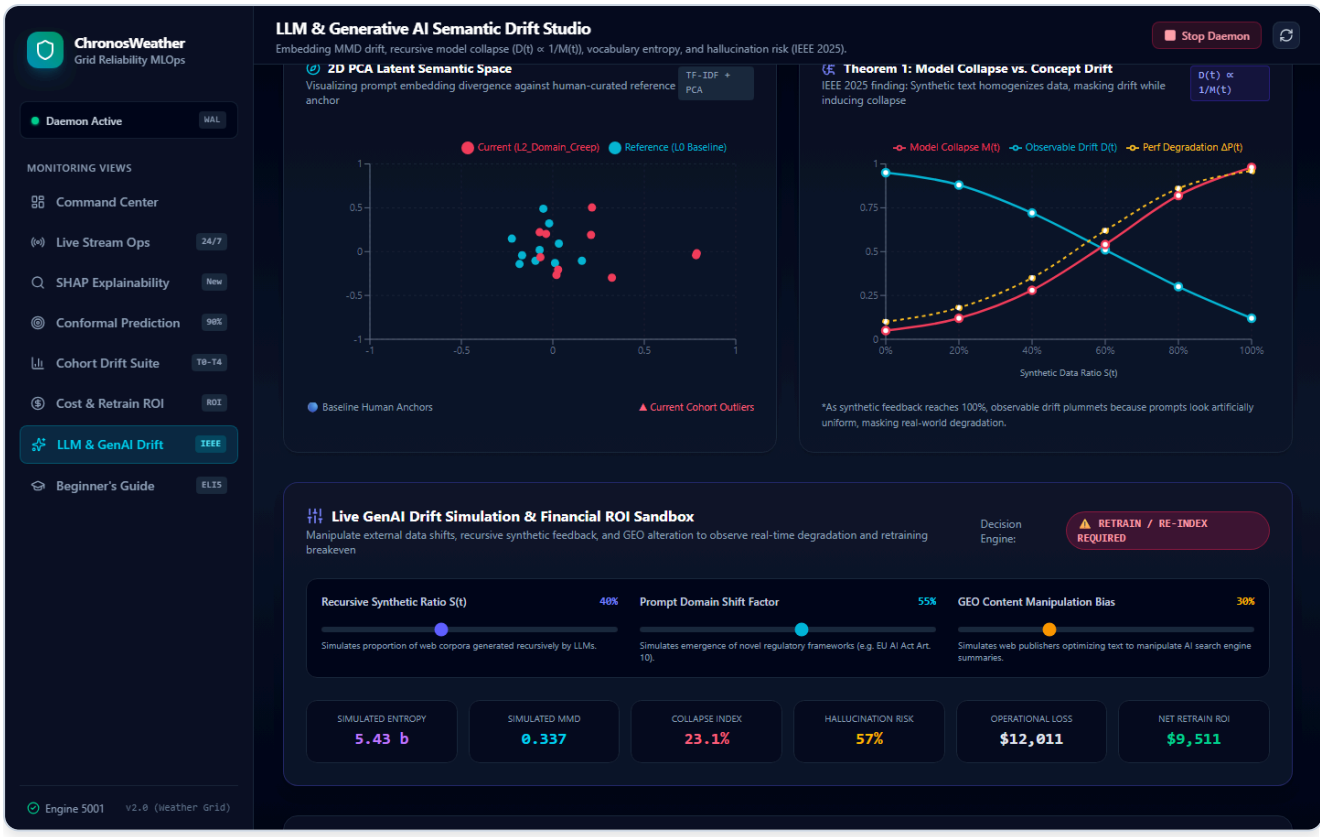


Figure 8: LLM & Generative AI Semantic Drift Studio (View #8) — IEEE 2025 Model Collapse Defense

The LLM Drift Studio extends surveillance into the generative AI frontier, implementing the IEEE 2025 research by Hajmohammed et al. The interface features a 2D PCA latent semantic scatter chart, the theoretical inverse law cross-over curve ($D(t) \propto 1/M(t)$), live prompt telemetry inspection, and interactive sliders for recursive synthetic data saturation $S(t)$, regulatory prompt shift, and Generative Engine Optimization (GEO) bias.

8.0 Limitations, Operational Boundaries & Future Scope

8.1 Theoretical & Practical Limitations

- **CBPE Calibration Invariance:** CBPE relies on the assumption that the model's calibration curve remains approximately preserved under pure covariate shift. Under extreme concept drift ($P(Y \mid X)$ fundamentally alters), CBPE can underestimate degradation. ChronosGuard mitigates this by fusing CBPE with conformal prediction set expansion.
- **Conformal Exchangeability:** The finite-sample 90% coverage theorem strictly requires exchangeability between calibration and test points. In time series with strong serial autocorrelation, coverage can momentarily fluctuate until rolling calibration windows update.
- **Local Embedding Dimensionality:** To ensure 100% free local execution without cloud API keys, the LLM module utilizes TF-IDF + PCA latent vector projections. While highly effective at capturing vocabulary entropy and semantic centroid shifts, production deployments with massive compute budgets can swap in open-source sentence-transformers (e.g., `all-MiniLM-L6-v2`).

8.2 Future Research Roadmap

1. **Online Continual Learning:** Extending sliding-window batch retraining into true streaming gradient updates via Online Passive-Aggressive algorithms or Elastic Weight Consolidation (EWC).
2. **Multi-Modal Drift Surveillance:** Expanding the MMD kernel suite to track joint image-text drift in Vision-Language Models (VLMs) used in autonomous driving and automated defect inspection.
3. **Distributed Cluster Streaming:** Migrating the single-node SQLite WAL architecture to distributed Apache Kafka + Ray streaming clusters for enterprise multi-datacenter deployments.

9.0 Academic References & Literature Citations

1. Hajmohammed, M., Chountas, P., & Chaussalet, T. (2025). *Concept Drift in Large Language Models: Challenges of Evolving Language, Contexts, and the Web*. In 2025 IEEE International Conference on Computational Intelligence and Advanced AI (ICCIAA), pp. 1–8. DOI: 10.1109/ICCIAA65327.2025.11013692.
2. Siciliani, T. (2024). *Drift Detection in Large Language Models: A Practical Guide*. Medium / Towards Data Science. URL: https://medium.com/@tsiciliani/drift-detection-in-large-language-models-a-practical-guide-3f54d783792c.
3. Shumailov, I., Shumaylov, Z., Zhao, Y., Gal, Y., Papernot, N., & Anderson, R. (2024). *The curse of recursion: Training on generated data makes models forget*. Nature, 631(8021), 755–759.
4. Vovk, V., Gammerman, A., & Shafer, G. (2005). *Algorithmic Learning in a Random World*. Springer Science & Business Media.
5. Lundberg, S. M., & Lee, S.-I. (2017). *A Unified Approach to Interpreting Model Predictions*. Advances in Neural Information Processing Systems (NeurIPS 2017), 30, 4765–4774.
6. Gretton, A., Borgwardt, K. M., Rasch, M. J., Schölkopf, B., & Smola, A. (2012). *A kernel two-sample test*. Journal of Machine Learning Research, 13(1), 723–773.
7. European Parliament and Council. (2024). *Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act)*, Article 10: Data and data governance. Official Journal of the European Union.
8. Zadrozny, B., & Elkan, C. (2002). *Transforming classifier scores into accurate multiclass probability estimates*. In Proceedings of the eighth ACM SIGKDD international conference on Knowledge discovery and data mining, pp. 694–699.

Appendix A: Comprehensive Technical Glossary

Term	Academic & Operational Definition
Covariate Shift	A distribution shift where the marginal probability distribution of inputs $P(X)$ changes between training and production while the conditional outcome distribution $P(Y \mid X)$ remains unchanged.
Concept Drift	A distribution shift where the statistical relationship between input features and target outcomes $P(Y \mid X)$ alters, regardless of whether marginal feature distributions $P(X)$ move.
Population Stability Index (PSI)	A binned symmetric divergence metric measuring proportional population shifts between a reference distribution and an operational batch.
Kolmogorov-Smirnov (KS) Test	A non-parametric two-sample statistical test evaluating the maximum vertical distance (D) between empirical cumulative distribution functions.
Wasserstein-1 Distance	The Earth Mover's Distance measuring the minimum total work required to transport one probability mass distribution into another.
Maximum Mean Discrepancy (MMD)	A non-parametric distance between probability measures computed via kernel embeddings in a Reproducing Kernel Hilbert Space (RKHS).
Confidence-Based Performance Estimation (CBPE)	An algorithmic framework that estimates production model error metrics (Accuracy, F1, ROC-AUC) directly from calibrated prediction probabilities without labels.
Expected Calibration Error (ECE)	The weighted absolute difference between model confidence and empirical accuracy across probability bins.
Split Conformal Prediction	A distribution-free uncertainty framework providing finite-sample valid prediction sets $C(X)$ satisfying $P(Y \in C(X)) \geq 1 - \alpha$.
Nonconformity Score (S_i)	A scalar metric quantifying how atypical or poorly predicted a candidate sample is relative to previous training instances.
Asymmetric Loss (C_{FN}, C_{FP})	An operational cost function where the penalty of a False Negative (e.g. blackout = $\$8,500$) vastly exceeds that of a False Positive (e.g. reservation = $\$350$).
TreeSHAP	A polynomial-time algorithm for computing exact Shapley values from tree ensemble models, evaluating feature attributions.
Directional Push	A SHAP metric indicating whether a temporal feature shift pushed predicted probabilities toward risk ($+1$) or safety (-1).
SQLite WAL Mode	Write-Ahead Logging architecture enabling concurrent lock-free reads while background threads append streaming records.
Model Collapse ($M(t)$)	Degenerative process where recursive training on synthetic AI data causes generative model distributions to lose variance and tail modes.
The Collapse Paradox	The counter-intuitive phenomenon where synthetic text homogenization causes observable drift ($D(t)$) to shrink while model utility severely collapses.
Shannon Vocabulary Entropy ($H(w)$)	Information-theoretic measure of token diversity in text queries, expressed in bits.

Term	Academic & Operational Definition
Type-Token Ratio (TTR)	The proportion of unique words $\|V\|$ relative to the total number of words N in a text corpus.
Generative Engine Optimization (GEO)	The practice of structuring web content specifically to manipulate LLM search engine citations and AI overview summaries.
Sliding-Window Reference Blending	A retraining strategy combining historical reference data with recent operational windows to adapt to drift without catastrophic forgetting.

Appendix B: Reproducibility & Local Execution Guide

B.1 System Requirements

- Operating System: Windows 10/11, Ubuntu 22.04+, or macOS Sonoma+.
- Python Runtime: Python 3.10+ (Developed and validated on Python 3.13.17).
- Node.js Runtime: Node.js 18+ and npm 9+ (for Vite frontend compilation).
- Hardware: Standard laptop / desktop (8GB RAM minimum; GPU not required).

B.2 Step-by-Step Launch Instructions

1. Launch Backend API Engine (Port 5001 / 5000)

```
# Navigate to project repository
cd "c:\Users\savin\Documents\01_Projects\Active\Capstone-Weather"

# Start Flask REST Engine (Default port 5001 for Weather; 5000 for FinTech)
python backend/api.py
```

2. Launch React 19 Frontend (Port 3001 / 3000)

```
# In a new terminal window
cd "c:\Users\savin\Documents\01_Projects\Active\Capstone-Weather\frontend"

# Start Vite Development Server
npm run dev -- --port 3001
```

3. Execute Complete Automated Test Suite (39 Tests)

```
# Run all unit and integration tests
cd "c:\Users\savin\Documents\01_Projects\Active\Capstone-Weather"
pytest tests/ -v
```

Expected output: `39 passed in ~16.77s` (Testing Conformal, Drift, Explainability, Streaming, Pipeline, and LLM Collapse).